

BIG DATA

PROGRAM STUDI INFORMATIKA /

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BAGIAN 1 — Langkah-Langkah Instalasi Hadoop

Ikuti setiap langkah di bawah ini dan sertakan screenshot sesuai ketentuan. Tempel screenshot pada area yang telah disediakan dan isi keterangan singkat.

Langkah 1 — Update & Upgrade System Packages

Jalankan perintah berikut untuk memastikan sistem Ubuntu 24.04 sudah up-to-date:

SS-1

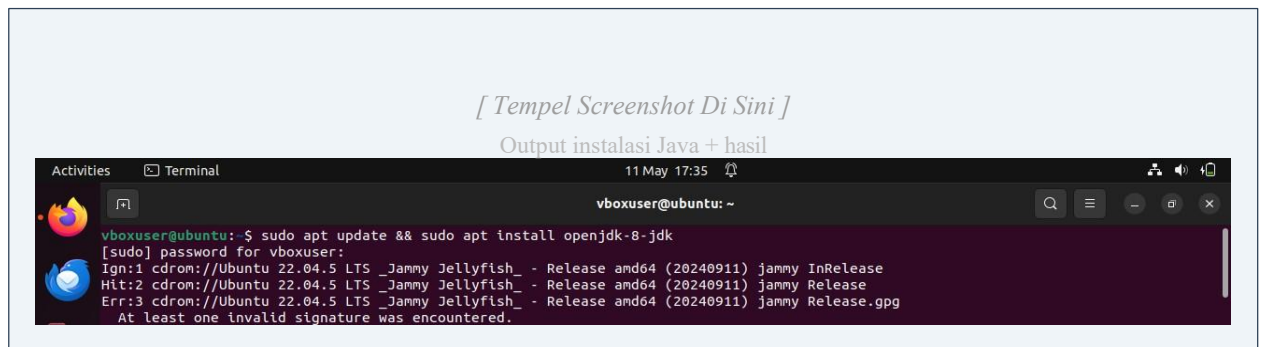


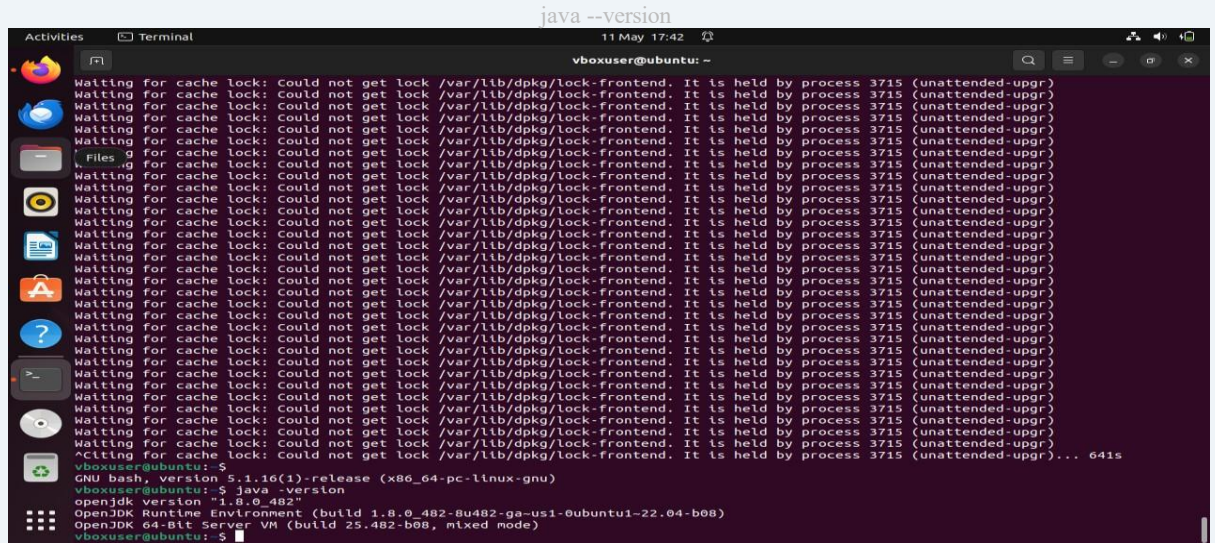
Gambar – SS-1

Langkah 2 — Instalasi Java (OpenJDK)

Hadoop memerlukan Java. Install OpenJDK dan verifikasi versinya:

SS-2



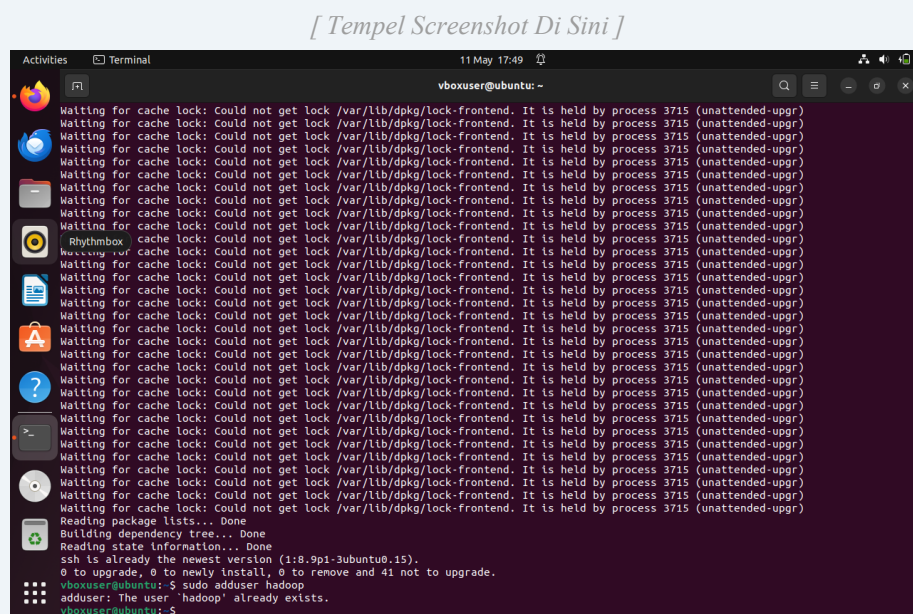


Gambar – SS-2

Langkah 3 — Membuat User Hadoop

Buat user khusus untuk Hadoop dan tambahkan ke grup sudo:

SS-3



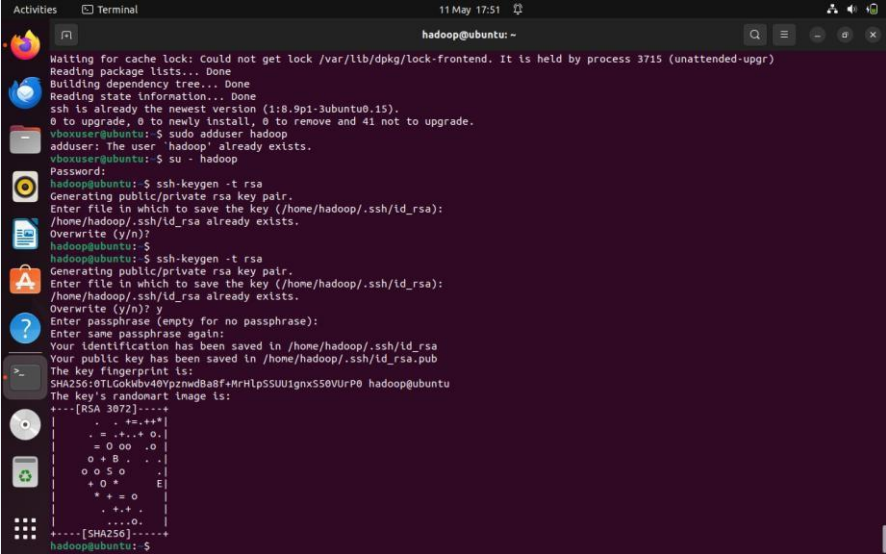
Output perintah adduser dan usermod

Gambar – SS-3

Langkah 4 — Konfigurasi SSH Passwordless

Generate SSH key agar node Hadoop dapat berkomunikasi tanpa password:

SS-4



```

Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 3715 (unattended-upgr)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ssh is already the newest version (1:8.9p1-3ubuntu0.15).
0 to upgrade, 0 to newly install, 0 to remove and 41 not to upgrade.
vboxuser@ubuntu:~$ sudo adduser hadoop
adduser: The user 'hadoop' already exists.
vboxuser@ubuntu:~$ su - hadoop
Password:
hadoop@ubuntu:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hadoop/.ssh/id_rsa):
/home/hadoop/.ssh/id_rsa already exists.
Overwrite (y/n)? y
hadoop@ubuntu:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hadoop/.ssh/id_rsa):
/home/hadoop/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/hadoop/.ssh/id_rsa
Your public key has been saved in /home/hadoop/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:0TLGokibv40YpznwDBaBfMrHlpSSUu1gnXS50VurP0 hadoop@ubuntu
The key's randomart image is:
+---[RSA 3072]----+
| . . +.+.+. |
| = .+.+.+. |
| = 0 00 . 0 |
| o + B . . . |
| o 0 5 0 . . |
| + 0 + . E |
| + + = 0 |
| .+.+. |
| ....0. |
+---[SHA256]-----+
hadoop@ubuntu:~$

```

[Tempel Screenshot Di Sini]

Output ssh-keygen (tampilkan fingerprint)

Gambar – SS-4

Langkah 5 — Download & Ekstrak Apache Hadoop

Download binary Hadoop dari server Apache dan ekstrak:

SS-5

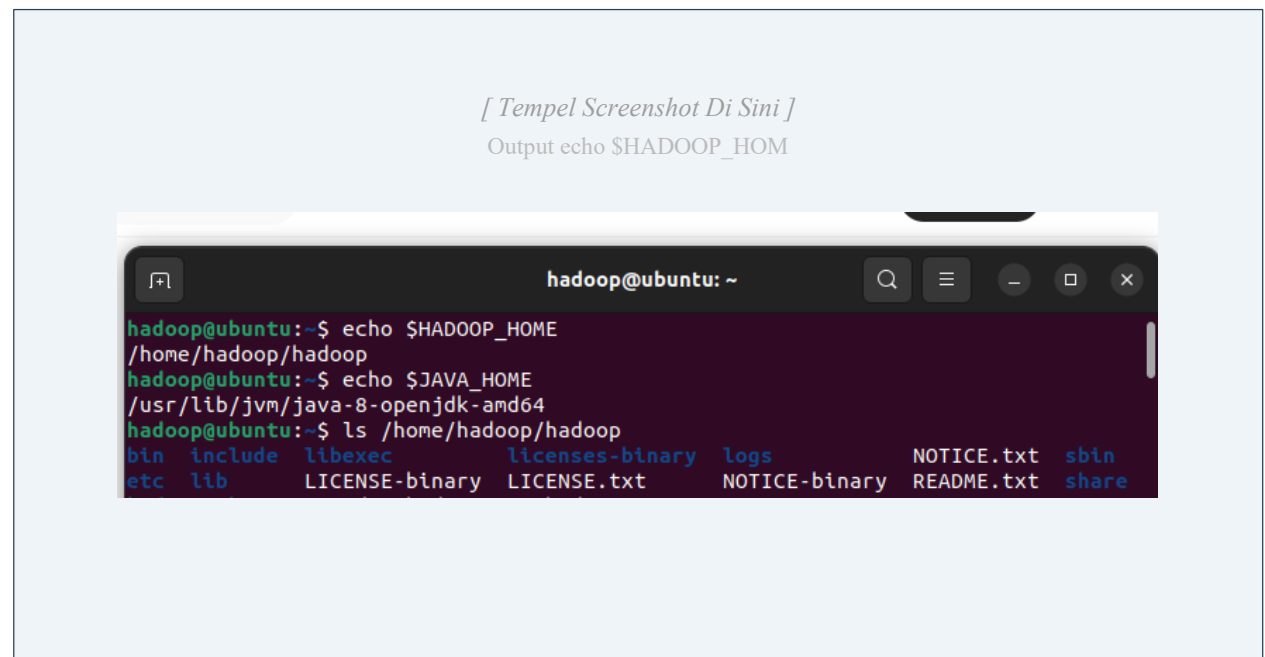


Gambar – SS-5

Langkah 6 — Konfigurasi Environment Variable

Set environment variable Hadoop di ~/.bashrc dan verifikasi:

SS-6



```

hadoop@ubuntu:~$ hadoop version
Hadoop 3.3.6
Repository: https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /home/hadoop/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
    
```

E dan hadoop version

Gambar – SS-6

Langkah 7 — Konfigurasi File XML Hadoop

Edit file konfigurasi XML (core-site.xml, hdfs-site.xml, mapred-site.xml, yarn-site.xml):

SS-7

```

GNU nano 6.2 /home/hadoop/hadoop/etc/hadoop/core-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<!-- Put site-specific property overrides in this file. -->
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
    
```

Tampilkan isi salah satu file XML (misal core-site.xml) di terminal

Gambar – SS-7

Langkah 8 — Format NameNode HDFS

Format NameNode untuk pertama kali sebelum menjalankan Hadoop:

- `hdfs namenode -format`

SS-8

```

hadoop@ubuntu:~/hadoop/logs$ nano -/hadoop/etc/hadoop/hdfs-site.xml
hadoop@ubuntu:~/hadoop/logs$ hdfs namenode -format
2026-05-11 20:07:34,240 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = ubuntu/127.0.1.1
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.3.6
STARTUP_MSG: classpath = /home/hadoop/hadoop/etc/hadoop:/home/hadoop/hadoop/share/hadoop/common/lib/netty-resolver-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-handler-ssl-ocsp-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar:/home/hadoop/hadoop/share/hadoop/common/lib/hadoop-auth-3.3.6.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-codec-nat-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jersey-json-1.20.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-native-epoll-4.1.89.Final-linux-aarch64.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-sctp-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-configuration2-2.8.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/reload4j-1.2.22.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jetty-utl-9.4.51.v20230217.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jzobjc-annotations-1.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerby-config-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jvoro-1.7.7.jar:/home/hadoop/hadoop/share/hadoop/common/lib/javax-servlet-api-3.1.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jackson-core-asl-1.9.13.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jackson-annotations-2.12.7.jar:/home/hadoop/hadoop/share/hadoop/common/lib/paranamer-2.3.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jersey-core-1.19.4.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jul-to-slf4j-1.7.36.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerb-core-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-lang3-3.12.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerby-pxt-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-codec-dns-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jsch-0.1.55.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerby-asn1-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/checker-qual-2.5.2.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-native-unix-common-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jersey-servlet-1.19.4.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-common-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jetty-servlet-9.4.51.v20230217.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-handler-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jackson-core-2.12.7.jar:/home/hadoop/hadoop/share/hadoop/common/lib/dnsjava-2.1.7.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-net-3.9.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jetty-io-9.4.51.v20230217.jar:/home/hadoop/hadoop/share/hadoop/common/lib/zipkin-3.6.3.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-native-queue-4.1.89.Final-osx-aarch64.jar:/home/hadoop/hadoop/share/hadoop/common/lib/zipkin-3.6.3.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-codec-socks-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-cli-1.2.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jaxb-impl-2.2.3-1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/token-provider-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/hadoop-shaded-proto-3.7.1-1.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-daemon-1.0.13.jar:/home/hadoop/hadoop/share/hadoop/common/lib/bouncycastle-1.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/hadoop-annotations-3.3.6.jar:/home/hadoop/hadoop/share/hadoop/common/lib/stax2-api-4.2.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerb-addr-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jaxb-api-2.2.11.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-native-queue-4.1.89.Final-osx-aarch64.jar:/home/hadoop/hadoop/share/hadoop/common/lib/httpclient-4.5.13.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerby-utl-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/audience-annotations-0.5.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-

```

Output hdfs namenode -format (bagian 'successfully formatted')

Gambar – SS-8

Langkah 9 — Start DFS & YARN + Verifikasi JPS

Jalankan layanan Hadoop dan verifikasi prosesnya:

SS-9

```

hadoop@ubuntu:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [ubuntu]
Starting resourcemanager
Starting nodemanagers
hadoop@ubuntu:~$ jps
53620 ResourceManager
53222 DataNode
54070 Jps
53414 SecondaryNameNode
53096 NameNode
53740 NodeManager
hadoop@ubuntu:~$

```

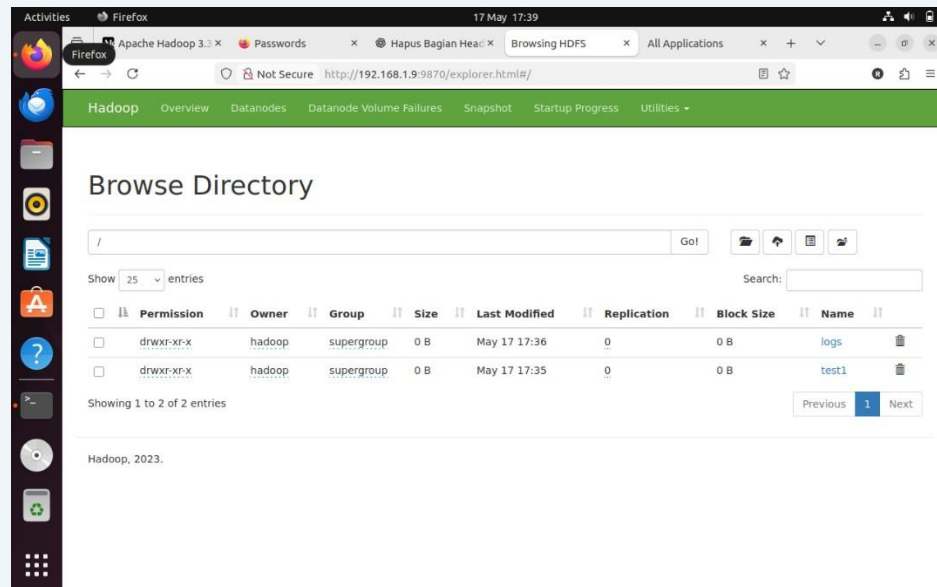
Output start-dfs.sh, start-yarn.sh, dan hasil perintah jps

Gambar – SS-9

Langkah 10 — Akses Web UI Hadoop (Namenode)

Buka browser dan akses Hadoop NameNode Web UI:

SS-10



Tampilan browser localhost:9870 (Hadoop NameNode Web UI)

Gambar – SS-10

Langkah 11 — Stop DFS & YARN + Verifikasi JPS

Hentikan layanan Hadoop setelah selesai digunakan:

SS-11

```
hadoop@ubuntu:~$ stop-all.sh
WARNING: Stopping all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: Use CTRL-C to abort.
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [ubuntu]
Stopping nodemanagers
Stopping resource manager
hadoop@ubuntu:~$
```

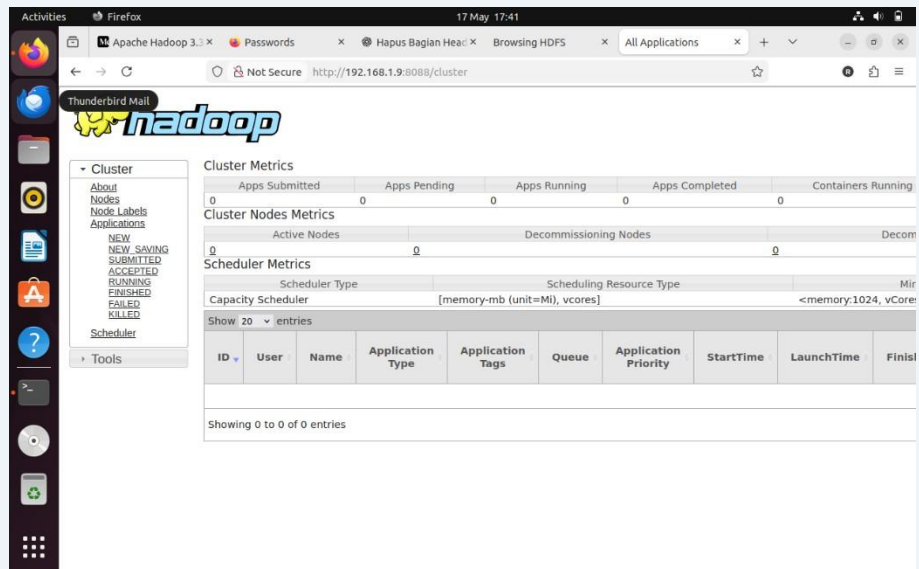
Output stop-dfs.sh, stop-yarn.sh, dan hasil jps setelah dihentikan

Gambar – SS-11

Langkah 12 (Optional) — Akses YARN Resource Manager

Akses YARN Resource Manager Web UI (optional, poin bonus):

SS-12 (Optional)



Tampilan browser localhost:8088 (YARN Resource Manager)

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